

Exploratory Notes on the Best-Bounded Inverse Submatrix Conjecture

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1 Problem

Let $U \in \mathbb{R}^{n \times k}$ have orthonormal columns. The Goreinov–Tyrtyshnikov–Zamarashkin hypothesis asks whether there is a $k \times k$ row submatrix Q with $\sigma_{\min}(Q) \geq n^{-1/2}$. This repository records exploratory work on the case $k = 3$ and related bounds; the unrestricted case remains open in these notes.

2 Status

The arguments and examples are provisional and are being discussed with researchers in matrix analysis. They should not be read as a finished theorem.

3 References

See `references/references.bib` and the accompanying open-access preprints.